How Good Does the Randomness Have to Be?

Noise quality and the reliability of feedforward probabilistic (p-bit) neural networks.

Working title (catchy): How Good Does the Randomness Have to Be? — Noise Quality and Reliability in Probabilistic Neural Networks

Alternatives:

- *When Sampling Breaks: Randomness-Quality Thresholds for p-bit Deep Networks*
- *Independent, Not Precise: Why Correlated Noise Breaks Probabilistic Neural Networks (leads with the headline finding)*
- *The Noise Makes the Neuron: Reliability of Configurable p-Neurons Under Imperfect Randomness*

Abstract

Probabilistic bits (p-bits) implement a neuron whose activation is produced by injected randomness: the neuron fires by comparing its input against a random number, so the *distribution* of that randomness sets the activation function (the inverse sampling theorem). Two recent works frame this project. Our instructors' group showed that decoupling a p-bit's noise path yields *configurable* activations (p-Tanh, p-Sigmoid, p-ReLU) selected by the injected noise distribution [1]. The Datta group showed that p-bit neurons work in *feedforward* deep networks (not just Boltzmann machines), that averaging a few stochastic samples at inference raises accuracy, and that this is energy-efficient [2]. Both assume **good** randomness. This project asks the question neither answers: **how good does the randomness have to be?** We build a feedforward p-bit network, inject controlled, hardware-motivated randomness defects (finite precision, bias, correlation, distribution mismatch), and measure where accuracy and — critically — the multi-sample accuracy benefit break down, using the inverse sampling theorem to predict the failures and an exact-ground-truth sampling task to prove them.

1. Background and Positioning

A p-neuron output is b = sign(f(I) − r), with r a random draw; it fires with probability equal to the CDF of r evaluated at f(I). Choosing the noise distribution therefore chooses the activation: uniform noise gives a ramp/ReLU-like curve, logistic noise gives the sigmoid, Gaussian noise gives the probit. Reference [1] realizes exactly this as hardware “modular” p-bits with configurable activations, but validates only single neurons and a 3-neuron logic gate — no trained network, no task. Reference [2] places p-neurons of the form b = sign(tanh(W) − r), r uniform, inside feedforward DNNs, trains them with a sample-aware straight-through estimator (or adds noise to a pre-trained deterministic model), and reports that a few samples per inference recover or exceed multi-bit deterministic accuracy on CIFAR-10, with an FPGA MNIST generator confirming the energy case. Neither paper studies degraded randomness. That gap — reliability of a feedforward p-DNN under imperfect randomness — is this project.

Prior work	p-neuron in feedforward NN?	Real task (MNIST/CIFAR)?	Configurable activation via noise?	Studies noise QUALITY / defects?
[1] Configurable p-Neurons (our instructors)	No (single neuron + logic gate)	No	Yes — core	No

Prior work	p-neuron in feedforward NN?	Real task (MNIST/CIFAR)?	Configurable activation via noise?	Studies noise QUALITY / defects?
[2] Datta / Ghantasala	Yes	Yes	No (fixed activation, uniform noise)	No (assumes ideal RNG)
THIS PROJECT	**Yes**	**Yes (MNIST)**	**Yes (IST lens)**	**Yes — the focus**

2. Research Questions and Hypotheses

The study is organized around four falsifiable hypotheses:

- **H1 (mechanism).** Each randomness defect deforms the p-neuron's activation curve exactly as its distorted CDF predicts: bias shifts it, finite precision turns it into stair-steps, a wrong distribution changes the activation family. Testable at the single-neuron level.
- **H2 (precision tolerance).** A feedforward p-DNN tolerates surprisingly low RNG precision — only a few bits — before accuracy drops, because averaging across many neurons and samples washes out quantization. (Good news for cheap hardware.)
- **H3 (correlation fragility — the headline).** The multi-sample accuracy benefit that makes p-DNNs attractive collapses under *correlated* randomness even when per-draw precision is adequate, because correlated samples are not independent draws. In short: for the sampling advantage, noise *independence* matters more than noise *precision*.
- **H4 (task dependence).** The same defect that barely dents classification accuracy pushes an exact-ground-truth sampling task measurably off its known target — noise sensitivity is strongly task-dependent.

3. Methodology (detailed)

3.1 The p-neuron primitive

Implement the p-neuron in NumPy/PyTorch as b = sign(f(I) – r), decomposed into three explicit sub-operations, following the hardware structure of [1] and [2]: (i) the deterministic pre-activation f(I); (ii) an RNG that draws r from a chosen distribution; (iii) a comparator/sign. Fresh r is drawn on every forward pass, so randomness is present at inference by construction. The injected-noise distribution is a first-class, swappable component — this is where every experiment acts.

3.2 Network and task

A small feedforward classifier on MNIST (downscaled or subset first for fast iteration), mirroring the architecture style of [2] (a compact convolutional or MLP network). MNIST classification is the primary testbed; accuracy and the multi-sample benefit are the main signals. Keep the network small and fixed across all noise conditions so that any change in behavior is attributable to the randomness, not the architecture.

3.3 Training strategy (isolating the variable of interest)

Two routes exist in [2]; we choose the one that isolates noise quality. **Primary route:** train a deterministic network normally (ordinary backprop), then at inference replace activations with p-neurons and inject the noise under test — the reference [2] “add-noise-to-a-trained-model” method. Because the weights are frozen, every accuracy change is caused by the injected randomness alone, which is exactly what a reliability study needs. **Optional extension** (time permitting): sample-aware training with a straight-through estimator, per [2], to test whether training *with* noise buys robustness to defective noise.

3.4 The noise-defect model (independent variables)

Four hardware-motivated defect families, each a knob we sweep. All are grounded in real devices: sMTJ p-bits produce only near-ideal, temperature-dependent noise [1], and FPGA implementations draw r from finite-width LFSRs [2].

- **Finite precision (bit-depth k).** Quantize r to k bits, sweeping k from 16 down to 1. Emulates a finite comparator / LFSR width.
- **Bias / offset.** Shift the mean of r. Emulates comparator offset or an asymmetric device.
- **Correlation.** Draw r from a short-period LFSR, or reuse/serially-correlate r across neurons and samples. Emulates a low-quality or shared RNG.
- **Distribution mismatch.** Inject a wrong-shaped noise (e.g. triangular or truncated) where uniform/logistic was intended, changing the activation family.

3.5 The IST lens: predict, then verify

For each defect, first *predict* the deformed activation from the distorted CDF (bias→horizontal shift; low precision→stair-steps; correlation→correct marginal but non-independent draws; wrong distribution→wrong curve). Then *measure* the actual activation of a single p-neuron under that defect and overlay prediction on measurement (tests H1). This converts “bad randomness” from a vague input into a specific, predicted mechanism — the intellectual spine inherited from the inverse sampling theorem and from [1].

3.6 Metrics and the exact-ground-truth harness

- **Classification accuracy** vs. defect strength (all four defects).
- **Multi-sample benefit**: accuracy vs. number of inference samples, for independent vs. correlated noise — the test of H3.
- **Calibration** (expected calibration error) of the multi-sample uncertainty estimates as noise quality degrades.
- **Single-neuron activation fidelity**: measured vs. IST-predicted curve per defect (H1).
- **Exact ground truth**: the frustrated-triangle p-bit sampler from Chapter 3 has a closed-form target distribution. Measure total-variation distance from that exact distribution vs. defect strength — a task where the deviation is *provable*, anchoring the softer MNIST metrics and testing H4.

4. Phased Plan (aligned to the course calendar)

Phase	Window	Key activities	Output / gate
P0 — Primitive & ground truth	Wk 1–2 (to Jul 9 p-bit lecture)	Implement the p-neuron; reproduce the frustrated-triangle sampler and validate against the closed-form distribution; reproduce a basic p-DNN on MNIST via the add-noise route [2].	Validated primitive + MNIST baseline running.
P1 — IST single-neuron study	Wk 2 (group research Jul 10)	Build the noise-defect library (precision, bias, correlation, mismatch); characterize each defect's activation deformation vs. IST prediction.	Defect library + activation-deformation figure (H1).
P2 — Reliability sweeps	Wk 3 (Jul 13–17)	Inject each defect into the MNIST p-DNN; sweep strength; produce accuracy-vs-quality curves and locate the failure knee.	Accuracy-vs-bit-depth & per-defect curves (H2).
P3 — Sampling fragility (headline)	Wk 3–4 (group research Jul 17)	Vary number of samples under independent vs. correlated noise; measure how defects erode the multi-sample benefit and calibration; run the exact-ground-truth cross-task comparison.	The headline plot (H3) + cross-task result (H4).
P4 — Write-up & capstone	Wk 4 (Jul 20–24)	Assemble paper and 10–13 min presentation; multiple seeds and error bars; honest reporting.	Final paper + capstone talk (Jul 23–24).

5. Results to Drive Toward

The project targets five concrete deliverable figures/claims:

- **Activation-deformation panel:** ideal vs. defective p-neuron activation, measured against the IST-predicted curve, for each defect.
- **Precision-tolerance curve:** MNIST accuracy vs. RNG bit-depth, with the failure knee marked (expected: tolerant down to a few bits).
- **Headline plot — sampling fragility:** accuracy vs. number of samples for independent vs. correlated noise, showing correlation flattens the sampling gain that is the whole point of a p-DNN. Target claim: *independence matters more than precision.*
- **Calibration curve:** uncertainty calibration (ECE) vs. noise quality, connecting reliability to trustworthy-AI framing.
- **Cross-task sensitivity:** total-variation distance from the exact frustrated-triangle distribution vs. defect strength, overlaid with the classification-accuracy sensitivity — the provable half of the story.

> **TARGET HEADLINE CLAIM**
>
> "A feedforward p-bit network tolerates coarse, low-precision randomness, but its multi-sample accuracy advantage collapses under correlated randomness — so for probabilistic neural networks, the independence of the noise matters more than its precision." Neither anchor paper [1,2] addresses this; it is provable on the exact-ground-truth task and measurable on MNIST.

6. Scope, Risks, and Honest Positioning

- **Novelty, stated plainly.** Building a p-DNN is done by [2]; configurable activations are done by [1]. The contribution here is the reliability-under-imperfect-randomness study neither performs, plus the correlation-vs-precision finding (H3). We claim that, not more.
- **Risk: robustness may be undramatic.** Averaging can wash out defects (H2). Mitigation: the correlation axis (H3) and the exact-ground-truth task (H4) are where effects should be sharp; lead with those, and report robustness honestly if that is what we find.
- **Scope control.** Primary route is add-noise-to-a-trained-model [2] (no fragile stochastic training); MNIST small first; one defect library reused across neuron, sampler, and classifier. Sample-aware training is an extension, not a dependency.

Deliverables: validated p-neuron + defect library (code); the five figures above; a research paper on the SRA template; a 10–13 minute capstone presentation.

> **TO CONFIRM WITH THE TA / INSTRUCTOR**
>
> - Primary training route: add-noise-to-a-trained-model, or sample-aware straight-through training? (We propose the former to isolate noise quality.)
> - Headline emphasis: the correlation-vs-precision finding (H3), or the cross-task exact-ground-truth reliability threshold (H4)?
> - Is a link to the configurable-activation idea from [1] (choosing the noise distribution) in scope, or is that reserved for their own follow-up?

References

[1] S. Bunaiyan, M. Alsharif, A. S. Abdelrahman, H. ElSawy, S. S. Cheema, S. A. Fahmy, K. Y. Camsari, F. Al-Dirini. “Configurable p-Neurons Using Modular p-Bits.” arXiv:2601.18943 (2026); IEEE ISCAS 2026.

[2] L. A. Ghantasala, M.-C. Li, R. Jaiswal, B. Behin-Aein, J. Makin, S. Sen, S. Datta. “Improving deep neural network performance through sampling.” arXiv:2507.07763 (2025); npj Unconventional Computing (2026).

[3] Course text, Chapter 3 (Discrete Optimization): p-bit sampling, the frustrated-triangle example, Ising models. INT 93LS, Summer 2026.

[4] K. Y. Camsari, R. Faria, B. M. Sutton, S. Datta. “Stochastic p-bits for invertible logic.” Phys. Rev. X 7, 031014 (2017).

[5] Related work (RBM activation deformation): Zeng et al., “Effect of stochastic activation function on reconstruction performance of RBMs with stochastic MTJs,” Appl. Phys. Lett. 124, 032404 (2024).

[6] Randomness-quality background (Project 1): PCG generator (O'Neill); A. Owen, Monte Carlo notes.

Working proposal for office-hours discussion. References [1] and [2] are the two anchor papers this project extends; [1] includes the course instructor and a TA as coauthors.